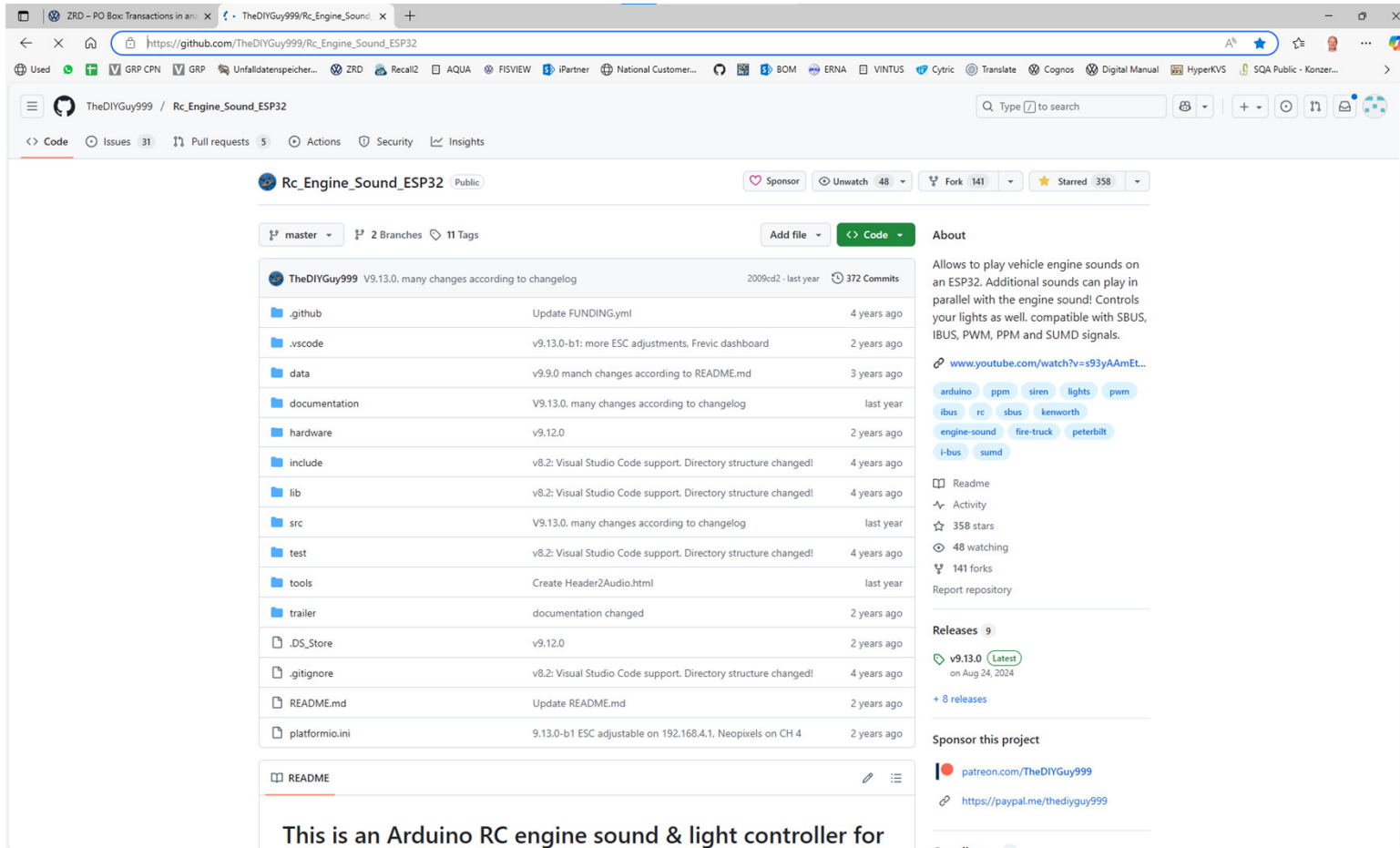


TheDIYGuy999/Rc_Engine_Sound_ESP32: Allows to play vehicle engine sounds on an ESP32. Additional sounds can play in parallel with the engine sound! Controls your lights as well. compatible with SBUS, IBUS, PWM, PPM and SUMD signals.



Browser tabs: ZRD - PO Box: Transactions in an... TheDIYGuy999/Rc_Engine_Sound x

Address bar: https://github.com/TheDIYGuy999/Rc_Engine_Sound_ESP32/tree/test

Navigation bar: TheDIYGuy999 / Rc_Engine_Sound_ESP32

Repository: Rc_Engine_Sound_ESP32 (Public)

Buttons: Sponsor, Unwatch 48, Fork 141, Starred 358

Branches: test (2 Branches), 11 Tags

Clone options: Local, Codespaces, Clone (HTTPS, SSH, GitHub CLI), Open with GitHub Desktop, Download ZIP

File list:

File/Folder	Commit Message	Time Ago
.github	Update FUNDING.y	
.vscode	v9.13.0-b1: more ES	
data	v9.9.0 manch chang	
documentation	9.14.0-b7: Changes	
hardware	v9.12.0	2 years ago
include	v8.2: Visual Studio Code support. Directory structure changed!	4 years ago
lib	v8.2: Visual Studio Code support. Directory structure changed!	4 years ago
src	9.14.0-b7: Changes according to Changelog.md	2 months ago
test	v8.2: Visual Studio Code support. Directory structure changed!	4 years ago
tools	9.14.0-b6: REVERSING_LOOP in the reversing beep file	5 months ago
trailer	documentation changed	2 years ago
.DS_Store	v9.12.0	2 years ago
.gitignore	v8.2: Visual Studio Code support. Directory structure changed!	4 years ago
README.md	9.14.0-b7: Changes according to Changelog.md	2 months ago
platformio.ini	9.14.0-b4: 7_Servos.h: SERVOS_HYDRAULIC_EXCAVATOR mode	6 months ago

README

URL: https://github.com/TheDIYGuy999/Rc_Engine_Sound_ESP32/archive/refs/heads/test.zip

About: Allows to play vehicle engine sounds on an ESP32. Additional sounds can play in parallel with the engine sound! Controls your lights as well. compatible with SBUS, IBUS, PWM, PPM and SUMD signals.

Links: www.youtube.com/watch?v=s93yAAMEt...

Tags: arduino, ppm, siren, lights, pwm, ibus, rc, sbus, kenworth, engine-sound, fire-truck, peterbilt, i-bus, sumd

Readme, Activity, 358 stars, 48 watching, 141 forks, Report repository

Releases 9: v9.13.0 (Latest) on Aug 24, 2024, + 8 releases

Sponsor this project: patreon.com/TheDIYGuy999, <https://paypal.me/thediuguy999>

MASTER

Does not
contain
Pingnon

Changelog

Todo:

- Remove in vehicle files: XENON_LIGHTS, LED_INDICATORS, INDICATOR_DIR, doubleFlashBlueLight
- Add to EEPROM & select box: DOUBLE_CLUTCH and other transmission options

9.13.0-b7:

- Defines removed: HAZARDS_WHILE_5TH_WHEEL_UNLOCKED
- Variables added (adjustable on 192.168.4.1): hazardsWhile5thWheelUnlocked
- Neopixel animation mode selectable in dropdown menu on 192.168.4.1
- Defines removed: NO_CABLIGHTS, NO_FOGLIGHTS, XENON_LIGHTS, FLICKERING_WHILE_CRANKING, SEPARATE_FULL_BEAM
- Variables removed: INDICATOR_DIR
- Variables added (adjustable on 192.168.4.1): noCabLights, noFogLights, xenonLights, flickeringWileCranking, swap_L_R_indicators, separateFullBeam, flashingBlueLight
- First light settings available on 192.168.4.1
- Website with collapsible sections to keep it more organized
- Servo positions adjustable on 192.168.4.1
- Improved website look
- Scania dashboard splash logo by Frikkiebester
- Dashboard requires less processing power, no engine rpm lagging anymore, if Frevic dashboard is used
- More adjustable ESC parameters on 192.168.4.1
- Dashboard improved & optional dashboard from Frevic implemented. Select it in "dashboard.h" by uncommenting "#define FREVIC_DASHBOARD"
- New parameter "#define NEOPIXEL_ON_CH4" in "6_Lights.h" allows to connect the NEOPIXELS to servo CH 4 (BUS mode only, no 5th wheel servo in this case)
- ESC parameters are adjustable via 192.168.4.1
- in "#define WEMOS_D1_MINI_ESP32" mode only: Experimental Serial command interface for ESC parameters

TEST

Files

test

Go to file

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.DS_Store

BigSurFix.md

Changelog.md

ESCLinearity.xlsx

Standard Parts List.pdf

Standard Parts List.xlsx

adjustmentsRemote.xlsx

gearRatios.xlsx

quickStartManual.pdf

servoCurves.xlsx

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10_Trailer.h

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2_Remote.h

Rc_Engine_Sound_ESP32 / documentation / Changelog.md

Preview Code Blame 731 lines (603 loc) · 44.5 KB

9.14.0-b7

- Hitachi ZW 370-6 wheel loader added
- FRSKY_TANDEM_HARMONY_LOADER remote profile added for TANDEM XE with touchscreen
- ESC settings added: HYDROSTATIC_MODE, directionChangeLimit
- Lights settings added: ROTATINGBEACON_ON_B1, INDICATOR_TOGGILING_MODE

9.14.0-b6

- Audio2Header.html now including "reversing beep" category
- REVERSING_LOOP in the reversing beep file allows to use sounds with different start beep (see CATreversingBeep.h)

9.14.0-b5

- Servo stroke limitation bug in EXCAVATOR_MODE fixed
- Hydraulic pump mixer added
- EEPROM only active, if ENABLE_WIRELESS is defined (configuration website is enabled). This allows to change default values (for example servo positions) without changing the EEPROM ID, if you don't want to use the website.
- Initial support for upcoming Pionon 14C excavator added

9.14.0-b4

- 7_Servos.h: SERVOS_HYDRAULIC_EXCAVATOR mode added. This allows to use non linear control curves for hydraulic valves, which eliminate the "dead zone" of most hydraulic valves. The curves are stored in curves.h. Also see servoCurves.xlsx

9.14.0-b3

- Volvo EC550EL excavator sounds added
- CH 1 - 4 servo outputs now usable for bucket, dipper, boom and swing in EXCAVATOR_MODE, including simulation of virtual inertia
- 2 position switch on CH 5 allows to switch between ISO and SAE control pattern without changing the wiring (swapping dipper and boom channels)

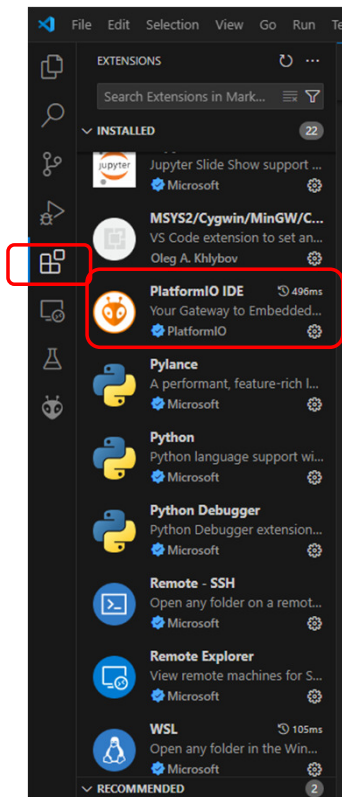
9.14.0-b2

- CAT 730 now with dump hydraulic sound

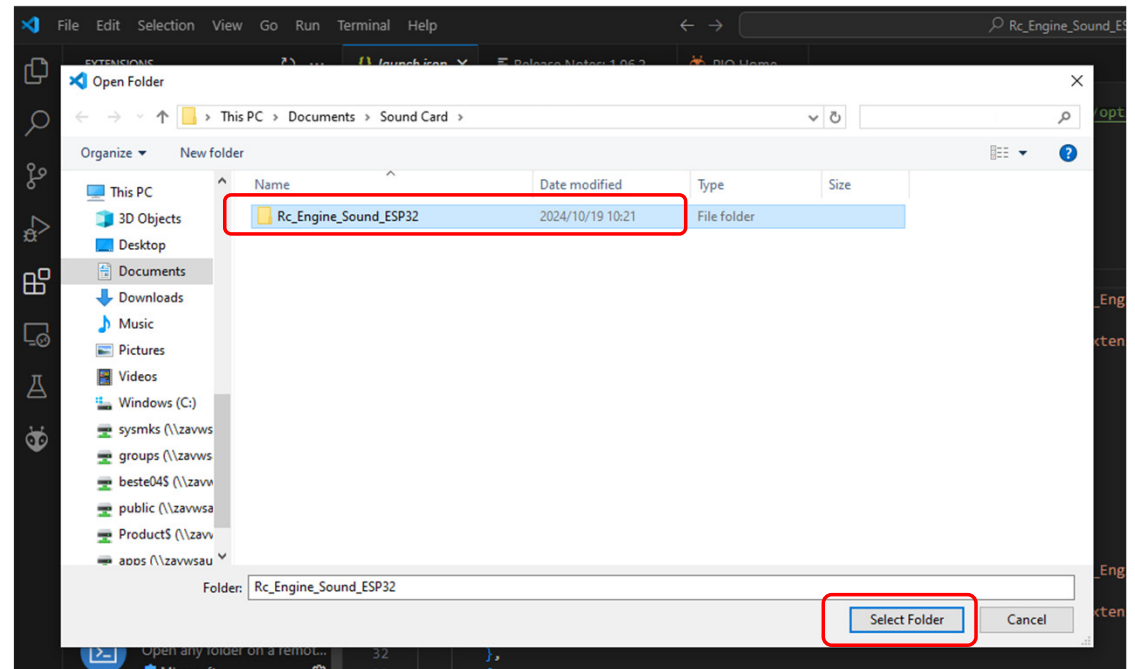
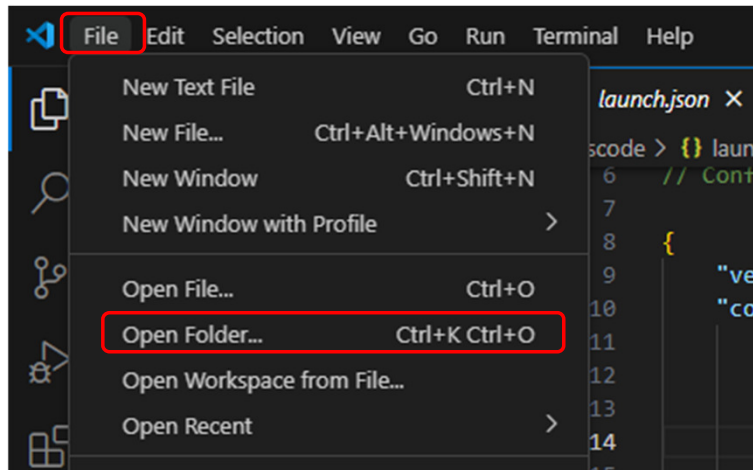
9.14.0-b1

- first CAT 730 implementation

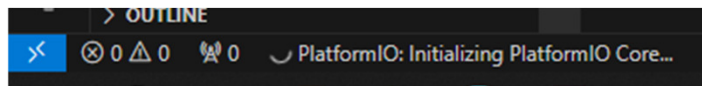
1. Install VSCode for you PC ([Download Visual Studio Code - Mac, Linux, Windows](#))
2. Accept and install ALL add-inns that VSCode requires.
3. Restart VSCode. Once open click on icon below and ensure PlatformIO IDE is installed



4. Open the downloaded folder from GITHUB



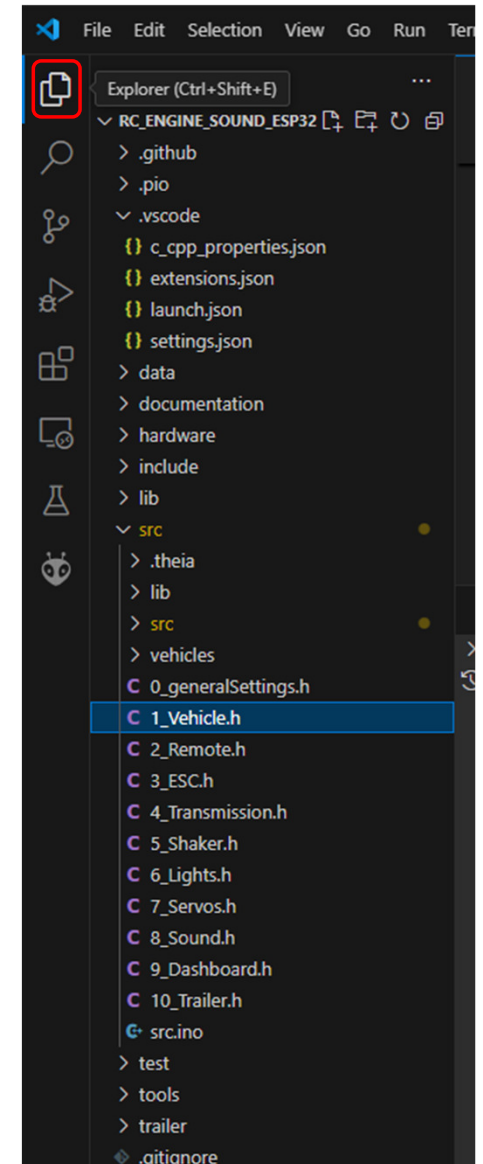
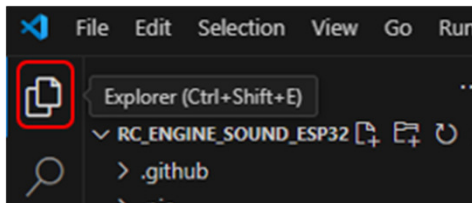
5. Allow the program to do what is required, accept and install any suggestions.
6. The first time it will run a bit longer to initialize everything



7. Once done you should have a bar like this at the bottom. The arrow is the upload button to the ESP32.



8. One more thing that you might need to install if you get any Squiggles errors is GIT. <https://git-scm.com/downloads/win>
9. To see all the folders you can click on icon.



RC_ENGINE_SOUND_ESP32-test

EXPLORER

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 - > lib
 - > src
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 - C 1_Vehicle.h
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 - C 3_ESC.h
 - C 4_Transmission.h
 - C 5_Shaker.h
 - C 6_Lights.h
 - C 7_Servos.h
 - C 8_Sound.h
 - C 9_Dashboard.h
 - C 10_Trailer.h
- > src.ino
- > test
- > tools
- > trailer
- > .gitignore
- > platformio.ini
- > README.md

src.ino C 0_generalSettings.h PIO Home platformio.ini

```
src > C 0_generalSettings.h > ...
1  #include <Arduino.h>
2  #include <WiFi.h>
3
4  /* GENERAL SETTINGS *****
5   * Have a look at the serial monitor in order to check the current settings of your vehicle!
6   * The required login informations for the browser based configuration via 192.186.4.1 can also be found there.
7   */
8
9  // Hardware settings -----
10 #define WEMOS_D1_MINI_ESP32 // Software is running on ESP32 RC Trailer controller, if defined (Headlights on GPIO 22 instead of 3, no cab lights support)
11
12 // Debug settings -----
13 // DEBUG options can slow down the playback loop! Only uncomment them for debugging, may slow down your system!
14 //#define DEBUG // More infos such as EEPROM dump on serial monitor, if defined
15 //#define CHANNEL_DEBUG // uncomment it for input signal & general debugging informations
16 //#define ESC_DEBUG // uncomment it to debug the ESC
17 //#define AUTO_TRANS_DEBUG // uncomment it to debug the automatic transmission
18 //#define MANUAL_TRANS_DEBUG // uncomment it to debug the manual transmission
19 //#define TRACKED_DEBUG // debugging tracked vehicle mode
20 //#define SERVO_DEBUG // uncomment it for servo calibration in BUS communication mode
21 //#define ESPNOW_DEBUG // uncomment for additional ESP-NOW messages
22 //#define CORE_DEBUG // Don't use this!
23
24 // EEPROM settings -----
25 uint8_t eeprom_id = 6; // change this id (between 1 and 255, compare with serial monitor), if you want to restore EEPROM defaults (executed if different) <----- NOTE!
26 // #define ERASE_EEPROM_ON_BOOT // EEPROM is completely overwritten, if defined! Never define it, vehicle will not work!
27 // only define it in order to clean up junk from old projects in your EEPROM
28
29 // Wireless settings -----
30 #define ENABLE_WIRELESS // Define this, if you want to use an ESP-Now wireless trailer or the WiFi configuration via 192.168.4.1
31
32 /* Wifi & ESP-Now and ESP-Now transmission power: less power = less speaker noise & longer battery life. Valid o
33 WIFI_POWER_19_5dBm = 78 // full power
34 WIFI_POWER_19dBm = 76
35 WIFI_POWER_18_5dBm = 74
36 WIFI_POWER_17dBm = 68
37 WIFI_POWER_15dBm = 60
38 WIFI_POWER_13dBm = 52
39 WIFI_POWER_11dBm = 44
40 WIFI_POWER_8_5dBm = 34
41 WIFI_POWER_7dBm = 28 // recommended setting
42 WIFI_POWER_5dBm = 20
43 WIFI_POWER_2dBm = 8 // lowest setting, WiFi may become weak
44 */
45 wifi_power_t cpType = WIFI_POWER_7dBm; // Only use values from above!
46
47 // Wifi settings (for vehicle configuration website, open 192.168.4.1 in your browser)-----
48 // Note: if these credentials were changed, using the configuration website,
49 // you can find the current ones in the serial monitor!
```

To use Trailer Card, uncomment (remove //)
#define WEMOS_D1_MINI_ESP32

To use WiFi Communications, uncomment
// #define ENABLE_WIRELESS

Ln 37, Col 22 Spaces: 4 UTF-8 LF () C++ Finish Setup PlatformIO

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C 5_Shaker.h

C 6_Lights.h

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Rc_Engine_Sound_ESP32-test

1_Vehicle.h

```
src > C 1_Vehicle.h
22 // EU trucks -----
36 // #include "vehicles/ManKat.h" // MAN KAT V8 Diesel German Bundeswehr military truck
37 // #include "vehicles/MagirusDeutz256.h" // Magirus Deutz 256 air cooled V8 Diesel truck
38 // #include "vehicles/MagirusMercur125.h" // Magirus Mercur air cooled V6 Diesel truck
39 // #include "vehicles/Saurer2DM.h" // Swiss Saurer 2DM R6 Diesel truck
40
41 // Russian trucks -----
42 // #include "vehicles/Ural4320.h" // URAL 4320 6x6 V8 Diesel military truck
43 // #include "vehicles/Ural375D.h" // URAL 375D 6x6 V8 petrol military truck
44 // #include "vehicles/URAL375.h" // URAL 375D 6x6 V8 petrol military truck (new version with better V8 sound, but good bass speaker required)
45 // #include "vehicles/GAZ66.h" // GAZ-66 V8 petrol military truck
46
47 // Russian tanks -----
48 // #include "vehicles/IS3.h" // IS-3 MW2 battle tank, V12 Diesel (dual ESC mode, good bass speaker required)
49
50 // Tractors -----
51 // #include "vehicles/KirovetsK700.h" // Russian Kirovets K700 monster tractor. Extreme turbo sound!
52 // #include "vehicles/FergusonTEA20.h" // 1940ies Ferguson TEA20 petrol engine tractor
53
54 // Excavators -----
55 // #include "vehicles/Volvo EC550ElExcavator.h" // Volvo EC550EL 55 ton excavator (use "FLYSKY_FS_I6S_EXCAVATOR" remote profile and SERVOS_EXCAVATOR servo profile)
56 // #include "vehicles/Caterpillar323Excavator.h" // Caterpillar 323 excavator (use "FLYSKY_FS_I6S_EXCAVATOR" remote profile)
57 // #include "vehicles/Pingon14CExcavator.h" // Pingon 14C excavator (use "FLYSKY_FS_I6S_EXCAVATOR" remote profile)
58 // Dumpers -----
59 // #include "vehicles/Benford3TonDumper.h" // Benford 3 ton dumper
60
61 // Wheel Loaders -----
62 // #include "vehicles/VolvoL120H.h"
63 // #include "vehicles/HitachiZW370.h" // Hitachi ZW 370-6 heavy wheel loader (for Lukas Cajkar Harmony 370 3D printer)
64
65 // Articulated Dump Trucks -----
66 // #include "vehicles/CAT730.h" // Caterpillar 730 heavy dump truck
67
68 // US motorcycles -----
69 // #include "vehicles/HarleyDavidsonFXSB.h" // Harley Davidson FXSB V2 motorcycle
70
71 // US cars -----
72 // #include "vehicles/ChevyNovaCoupeV8.h" // 1975 Chevy Nova Coupe V8
73 // #include "vehicles/1965FordMustangV8.h" // 1965 Ford Mustang V8
74 // #include "vehicles/Chevy468.h" // Chevy 468 big block V8
75
76 // EU cars -----
77 // #include "vehicles/VwBeetle.h" // VW Käfer / Beetle
78 // #include "vehicles/JaguarXJS.h" // Jaguar XJS V12, manual transmission
79 // #include "vehicles/JaguarXJSAutomatic.h" // Jaguar XJS V12, automatic transmission
80 // #include "vehicles/MGBGtV8.h" // MGB GT V8, manual transmission
81 // #include "vehicles/LaFerrari.h" // Ferrari LaFerrari, V12, 6 speed double clutch transmission
82
83 // US SUV & Pickups -----
```

Ln 57, Col 113 Spaces: 4 UTF-8 LF () C++ Finish Setup PlatformIO

Uncomment vehicle you want, ensure only one is uncommented

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C_4_Transmission.h

C_5_Shaker.h

C_6_Lights.h

C_7_Servos.h

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C_1_Vehicle.h

C_2_Remote.h

FLYSKY_FS_I6S_EXCAVATOR

```
1 #include <Arduino.h>
2
3 // Select (remove //) the remote configuration profile you have:
4 // #define FLYSKY_FS_I6X // <----- Flysky FS-16x
5 // #define FLYSKY_FS_I6S // <----- Flysky FS-16s
6 // #define FLYSKY_FS_I6S_LOADER // <----- Flysky FS-16s for BURNIE222 Volvo L120H loader (use IBUS communication setting)
7 #define FLYSKY_FS_I6S_EXCAVATOR // <----- Flysky FS-16s for KABOLITE K336 hydraulic excavator (use IBUS communication setting)
8 // #define FRISKY_TANDEM_EXCAVATOR // <----- Frsky Tandem XE for hydraulic excavator (use SBUS communication setting)
9 // #define FRISKY_TANDEM_HARMONY_LOADER // <----- Frsky Tandem XE for Lukas Cajkar Harmony 370 (use SBUS communication setting)
10 // #define FLYSKY_GT5 // <----- Flysky GT5 / Reely GT6 EVO / Absima CR6P
11 // #define RGT_EX86100 // <----- MT-305 remote delivered with RGT EX86100 crawler (use PWM communication setting)
12 // #define GRAUPNER_MZ_12 // <----- Graupner MZ-12 PRO
13 // #define MICRO_RC // <----- The car style DIY "Micro RC" remote. Don't use this with standard receiver
14 // #define MICRO_RC_STICK // <----- The stick based DIY "Micro RC" remote. Don't use this with standard receiver
15
16 // For testing only!
17 // #define FLYSKY_FS_I6S_EXCAVATOR_TEST // <----- Flysky FS-16s for KABOLITE K336 hydraulic excavator (use IBUS communication setting)
18
19 // BOARD SETTINGS *****
20 // Choose the board version
21 // #define PROTOTYPE_36 // 36 or 30 pin board (do not uncomment it or it will cause boot failure)
22
23 // COMMUNICATION SETTINGS *****
24 // Choose the receiver communication mode (never uncomment more than one!) !!! ADJUST YOUR RECEIVER SETTINGS (see the PIN LIST in README for more details)
25
26 // PWM servo signal communication (CH1 - CH6 headers, 6 channels) -----
27 // PWM mode active, if SBUS, IBUS, SUMD and PPM are disabled (// in front of #define)
28
29 // SBUS communication (RX header, 13 channels. This is my preferred communication profile)
30 // #define SBUS_COMMUNICATION // control signals are coming in via the SBUS interface
31 // NOTE: "boolean sbusInverted = true / false" was moved to the remote configuration
32 uint32_t sbusBaud = 100000; // Standard is 100000. Try to lower it, if your receiver has problems
33 #define EMBEDDED_SBUS // Embedded SBUS code is used instead of SBUS library
34 uint16_t sbusFailsafeTimeout = 100; // Failsafe is triggered after this timeout in milliseconds
35
36 // IBUS communication (RX header, 13 channels not recommended, NO FAILSAFE, if bad connection)
37 #define IBUS_COMMUNICATION // control signals are coming in via the IBUS interface
38
39 // SUMD communication (RX header, 12 channels, For Graupner remotes) -----
40 // #define SUMD_COMMUNICATION // control signals are coming in via the SUMD interface
41
42 // PPM communication (RX header, 8 channels, working fine, but channel signals are a bit noisy)
43 // #define PPM_COMMUNICATION // control signals are coming in via the PPM interface
44
45 // CHANNEL LINEARITY SETTINGS *****
46 // Note: avoid these options for excavators!
47 // #define EXPONENTIAL_THROTTLE // Exponential throttle curve. Ideal for enhanced slow speed control in crawlers
48 // #define EXPONENTIAL_STEERING // Exponential steering curve. More steering accuracy around center position
49
```

RC_ENGINE_SOUND_ESP32-test

C_1_Vehicle.h

C_2_Remote.h

```
1 #define FLYSKY_FS_I6S_LOADER
2
3 // Flysky FS-16s remote configuration profile (for excavators only)
4 #define FLYSKY_FS_I6S_EXCAVATOR
5
6 // NOTE: The vehicle file needs to contain #define EXCAVATOR_MODE
7
8 // Channel assignment (use NONE for non existing channels)
9 // Remote channel remapping // Sound controller channel remapping
10 #define STEERING 1 // CH1 bucket
11 #define GEARBOX 2 // CH2 dipper
12 #define THROTTLE 9 // CH3 1 position switch off, idle, rev. 9 (4 for test)
13 #define HORN 10 // CH4 horn
14 #define FUNCTION_R 3 // CH5 boom
15 #define FUNCTION_L 6 // CH6 left track 6 (none for test)
16 #define POT2 7 // CH7 right track
17 #define MODE1 4 // CH8 mode
18 #define MODE2 8 // CH9 light switch
19 #define MOMENTARY1 5 // CH10 2 position switch for ISO / SAE mode selection
20 #define HAZARDS NONE // CH11
21 #define INDICATOR_LEFT NONE // CH12
22 #define INDICATOR_RIGHT NONE // CH13
23
24 // Channels reversed or not
25 boolean channelReversed[14] = {
26 false, // CH0 (unused)
27 false, // CH1 (unused)
28 false, // CH2 (unused)
29 false, // CH3
30 false, // CH4
31 false, // CH5
32 false, // CH6
33 false, // CH7
34 false, // CH8
35 false, // CH9
36 false, // CH10
37 false, // CH11
38 false, // CH12
39 false, // CH13
40 };
41
42 // Channels auto zero adjustment or not (don't use it for channels without spring centered neutral position, switches or unused channels)
43 boolean channelAutoZero[14] = {
44 false, // CH0 (unused)
45 false, // CH1 (unused)
46 true, // CH2
47 true, // CH3
48 true, // CH4
49 false, // CH5
50 false, // CH6
51 false, // CH7
52 false, // CH8
53 false, // CH9
54 false, // CH10
55 false, // CH11
56 false, // CH12
57 false, // CH13
58 };
59
```

Ln 7, Col 130

Spaces: 4

UTF-8

LF

C++

Finish Setup

PlatformIO

Uncomment remote you want, ensure only one is uncommented. Scroll down for channel settings



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src > C_3_ESC.h > ...

```
115 // Additional reverse speed (disconnect & reconnect battery after changing this setting):
116 // - Hobbywing 1000 ESC & 551 340 motor for Hercules Hobby Trucks with 3 speed transmission = 80
121 // - Meccano Dumper = 0
122 // - RZ7886 Driver = 0
123 uint16_t escReversePlus = 0;
124
125 // Crawler mode escRampTime (see "8_Sound.h") WARNING: a very low setting may damage your transmission!
126 uint8_t crawlerEscRampTime = 10; // about 10 (15 for Jeep), less = more direct control = less virtual inertia
127
128 // Allows to scale vehicle file dependent acceleration
129 uint16_t globalAccelerationPercentage = 100; // about 100 - 200% (200 for Jeep, 150 for 1/8 Landy) Experimental, may cause automatic transmission shifting issues!
130
131 // BATTERY PROTECTION SETTINGS *****
132
133 /* Battery low discharge protection (only for boards with voltage divider resistors):
134 * IMPORTANT: Enter used resistor values in Ohms (Ω) and THEN adjust DIODE_DROP, until your readings match the actual battery voltage! */
135 // #define BATTERY_PROTECTION // This will disable the ESC output, if the battery cutout voltage is reached. 2 fast flashes = battery error!
136 const float CUTOFF_VOLTAGE = 3.3; // Usually 3.3 V per LiPo cell. NEVER below 3.2 V!
137 const float FULLY_CHARGED_VOLTAGE = 4.2; // Usually 4.2 V per LiPo cell, NEVER above!
138 const float RECOVERY_HYSTERESIS = 0.2; // around 0.2 V
139 /* Note on resistor values: These values will be used to calculate the actual ratio between these two resistors
140 * When selecting resistors, always use two of the same magnitude: Like, for example, 10k/2k, 20k/4k or 100k/20k
141 * WARNING: If the ratio is too LOW, like 10k/5k (2:1 = 2), the battery voltage will most likely DAMAGE the con
142 * Example calculation: 2000 / (2000 + 10000) = 0.166 666 666 7; 7.4 V * 0.167 = 1.2358 V (of 3.3 V maximum on GPIO Pin). */
143 uint32_t RESISTOR_TO_BATTERY_PLUS = 9400; // Value in Ohms (Ω), for example 10000
144 uint32_t RESISTOR_TO_GND = 1000; // Value in Ohms (Ω), for example 2000. Measuring exact resistor values before soldering, if possible is recommended!
145 float DIODE_DROP = 0.31; // Fine adjust measured value and/or consider diode voltage drop (about 0.34V for SS34 diode)
146 /* It is recommended to add a sticker to your ESP32, which includes the 3 calibration values above */
147 volatile int outOfFuelVolumePercentage = 80; // Adjust the message volume in %
148 // Select the out of fuel message you want:
149 #include "vehicles/sounds/OutOfFuelEnglish.h"
150 // #include "vehicles/sounds/OutOfFuelGerman.h"
151 // #include "vehicles/sounds/OutOfFuelFrench.h"
152 // #include "vehicles/sounds/OutOfFuelDutch.h"
153 // #include "vehicles/sounds/OutOfFuelSpanish.h"
154 // #include "vehicles/sounds/OutOfFuelPortuguese.h"
155 // #include "vehicles/sounds/OutOfFuelJapanese.h"
156 // #include "vehicles/sounds/OutOfFuelChinese.h"
157 // #include "vehicles/sounds/OutOfFuelTurkish.h"
158 // #include "vehicles/sounds/OutOfFuelRussian.h"
159
```

If Battery Protection resistors are not fitted, ensure this is commented out.

Ln 135, Col 155 Spaces: 2 UTF-8 LF {} C++ Finish Setup PlatformIO